Data Analytics (DA)

MySQL Assignment 2 - Querying Data

Objective

The focus of this assignment is on data insertion, querying, filtering, aggregation, and result interpretation while ensuring consistency with the existing database design.

Step1: Inserted Department values

```
INSERT INTO Departments VALUES
(1,'Finance'),
(2,'IT'),
(3,'HR'),
(4,'Marketing');
select * from Departments;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

department_id	department_name
1	Finance
3	HR
2	IT
4	Marketing
NULL	NULL

Tasks:

Step2: Distinct values

```
select distinct salary from Employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

salary
50000
55000
70000
52000

Step3: Alias Query

```
SELECT age AS Employee_Age,
salary AS Employee_Salary
FROM Employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Employee_Age	Employee_Salary
19	50000
20	55000
21	70000
22	52000

Step4: where condition

```
SELECT *
FROM Employees
WHERE salary > 50000
AND hire_date < '2016-01-01';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	employee_name	age	gender	designation	hire_date	department_id	location_id	salary
3	Novak	21	M	Hr	2014-11-20	1	3	70000
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Step5: missing values filled with data scientist

```
SELECT *
FROM Employees
WHERE salary > 50000
AND hire_date < '2016-01-01';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	employee_name	age	gender	designation	hire_date	department_id	location_id	salary
3	Novak	21	M	Hr	2014-11-20	1	3	70000
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

No missing values found as null to update

Step6: Sorting

```
SELECT *
FROM Employees
ORDER BY department_id ASC, salary DESC;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	employee_name	age	gender	designation	hire_date	department_id	location_id	salary
3	Novak	21	M	Hr	2014-11-20	1	3	70000
1	Roger	19	M	Player	2015-05-10	1	1	50000
2	Sharmila	20	F	Analyst	2016-03-12	2	2	55000
4	Suchitra	22	F	Manager	2019-06-15	3	1	52000
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Step7: limit

```
SELECT *
SELECT *
FROM Employees
WHERE YEAR(hire_date) = 2018
LIMIT 5;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	employee_name	age	gender	designation	hire_date	department_id	location_id	salary
5	Vaishnavi	26	M	Analyst	2018-07-10	2	2	58000
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Step8; min()

```
select MIN(age)
FROM Employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

MIN(age)
19

Step9: maximum salary

```
select location_id, max(Salary) from Employees group by location_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

location_id	max(Salary)
1	52000
2	58000
3	70000

Step10: Calculate the average salary for each designation containing the word 'Analyst'.

```
select location_id, avg(salary) from Employees where designation = "Analyst" group by location_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

location_id	avg(salary)
2	56500.0000

Step11: Calculate the sum of all salaries in the Finance department.

```
SELECT SUM(salary)
FROM Employees
WHERE department_id = 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

SUM(salary)
120000

Step12: Join

```
SELECT e.employee_name, e.designation, d.department_name
FROM Employees e
INNER JOIN Departments d
ON e.department_id = d.department_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

employee_name	designation	department_name
Roger	Player	Finance
Novak	Hr	Finance
Suchitra	Manager	HR
Sharmila	Analyst	IT
Vaishnavi	Analyst	IT

Step13: Left Join

```
SELECT d.department_name, COUNT(e.employee_id)
FROM Departments d
LEFT JOIN Employees e
ON d.department_id = e.department_id
GROUP BY d.department_name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

department_name	COUNT(e.employee_id)
Finance	2
HR	1
IT	2
Marketing	0

Step14: Right join

```
SELECT l.location_name, e.employee_name
FROM Employees e
RIGHT JOIN Location l
ON e.location_id = l.location_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

location_name	employee_name
Bangalore	Sharmila
Bangalore	Vaishnavi
Chennai	Roger
Chennai	Suchitra
Hyderabad	Novak

Step15: Cross Join

```
SELECT Departments.department_name, Location.location_name
FROM Departments
CROSS JOIN Location;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

department_name	location_name
Finance	Bangalore
HR	Hyderabad
HR	Chennai
HR	Bangalore
IT	Hyderabad
IT	Chennai
IT	Bangalore
Marketing	Hyderabad
Marketing	Chennai
Marketing	Bangalore

Step16: Self Join

```
SELECT e1.employee_name, e2.employee_name
FROM Employees e1
JOIN Employees e2
ON e1.department_id = e2.department_id
AND e1.employee_id <> e2.employee_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

employee_name	employee_name
Novak	Roger
Vaishnavi	Sharmila
Roger	Novak
Sharmila	Vaishnavi

Windows Functions:

Write a window function query to rank employees by salary using rank().

```
SELECT employee_name, salary,
RANK() OVER (ORDER BY salary DESC) AS salary_rank
FROM Employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

employee_name	salary	salary_rank
Novak	70000	1
Vaishnavi	58000	2
Sharmila	55000	3
Suchitra	52000	4
Roger	50000	5

Write a window function query to rank employees by salary within each department using DENSE_RANK()

```
SELECT employee_name, department_id, salary,
DENSE_RANK() OVER (PARTITION BY department_id ORDER BY salary DESC) AS dept_rank
FROM Employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

employee_name	department_id	salary	dept_rank
Novak	1	70000	1
Roger	1	50000	2
Vaishnavi	2	58000	1
Sharmila	2	55000	2
Suchitra	3	52000	1

Write a window function query, Running total salary by department

```
SELECT employee_name, department_id, salary,
SUM(salary) OVER (PARTITION BY department_id ORDER BY salary) AS running_total
FROM Employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

employee_name	department_id	salary	running_total
Roger	1	50000	50000
Novak	1	70000	120000
Sharmila	2	55000	55000
Vaishnavi	2	58000	113000
Suchitra	3	52000	52000